

Anti-obesity and antioxidative effects of purple sweet potato extract in 3T3-L1 adipocytes in vitro.



The purpose of the current study was to determine the anti-obesity and anti-inflammatory effects of an extract of purple sweet potatoes (PSPs) on 3T3-L1 adipocytes. For this purpose, differentiated 3T3-L1 adipocytes were treated with a PSP extract at concentrations of 1,000, 2,000, and 3,000 $\mu\text{g/mL}$ for 24 hours. Then, we measured the changes in the sizes of the adipocytes, the secretion of leptin, and the mRNA/protein expression of lipogenic, inflammatory, and lipolytic factors after the treatment with the PSP extract. The PSP extract diminished leptin secretion, indicating that growth of fat droplets was suppressed. The extract also suppressed the expression of mRNAs of lipogenic and inflammatory factors and promoted lipolytic action. The antioxidative activity of the PSP extract was also measured using three different in vitro methods: 1,1-diphenyl-2-picrylhydrazyl free radical scavenging activity, ferric reducing ability potential assay, and chelating activity of transition metal ions. Taken together, our study shows that PSP extract has antilipogenic, anti-inflammatory, and lipolytic effects on adipocytes and has radical scavenging and reducing activity.